ENGINEERING JUDGMENT SUMMARY

CareerForge Day 5: AI Is Not The Engineer

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GITHUB: https://github.com/liacodes49/Career-Forge/tree/day-05-ai-is-not-the-engineer

WHY I DID NOT BLINDLY FOLLOW THE AI

AI systems produced technically plausible recommendations. However, plausibility is not proof. Every recommendation was evaluated on three criteria:

- Is the evidence behind this claim independently verifiable?
- Does this action carry reversible or irreversible consequences?
- What is the failure mode if the AI's assumption is wrong?

WHAT I TRUSTED

The root cause diagnosis (memory leak / OOMKilled) was supported by observable data,pod logs and Prometheus alerts are independent sources. This was treated as a credible hypothesis, not a confirmed fact, but worth acting on as a first investigation step.

The maxPoolSize configuration recommendation is consistent with well-established DB connection management patterns. The value of 10 is a reasonable starting point, not inherently dangerous, and testable in staging.

WHAT I MODIFIED

The pod restart was accepted as the correct immediate action, but only after adding the pre-condition of draining in-flight transactions. An AI system cannot know the current transaction state. Restarting a payments service with active transactions is a data integrity risk that the AI did not account for.

The memory limit increase from 512Mi to 2Gi was modified to a deferred action. Before increasing limits, node capacity must be confirmed. An unverified limit increase could trigger host-level OOM, taking down other pods on the same node, a wider blast radius than the original incident.

WHAT I BLOCKED

Two recommendations were blocked entirely:

- Deploying the DB config changes directly to production without staging. There is no engineering justification for skipping staging on a database configuration change. The AI offered no evidence that this shortcut is safe.
- The 'zero blast radius' claim. This is the most dangerous statement in the AI's output. Payments services are never isolated. They have upstream callers, downstream ledger systems, and async retry queues. Declaring zero risk without a dependency map is not analysis, it is false confidence.

CORE ENGINEERING PRINCIPLE APPLIED

Irreversible decisions require higher evidence bars than reversible ones.

- Restarting a pod = reversible → proceed with precautions
- Skipping staging for a DB change = effectively irreversible in prod → block
- Declaring zero blast radius without evidence = potentially irreversible if wrong → block

AI accelerates diagnosis. Engineers own decisions.

WHAT GOOD AI-ASSISTED ENGINEERING LOOKS LIKE

The correct workflow is:

- Use AI to generate hypotheses and flag patterns
- Independently verify each claim with primary evidence (logs, metrics, traces)
- Apply engineering judgment to classify: Trust / Modify / Block
- Document your reasoning, not just the action
- Own the final decision; **AI cannot be accountable, you can.**